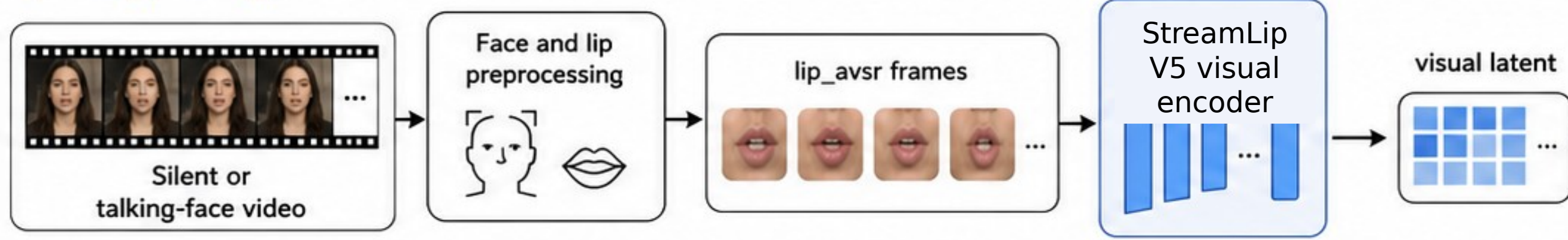


Audio-Centric Visual Speech Reconstruction

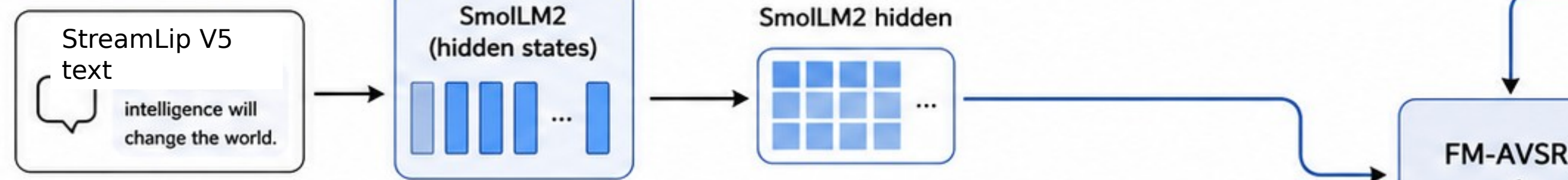
Changxun Pan, Zihan Guo, Anqiao Fu

Generate speech by reconstructing Mimi audio latents, not by cascading lip → text → TTS

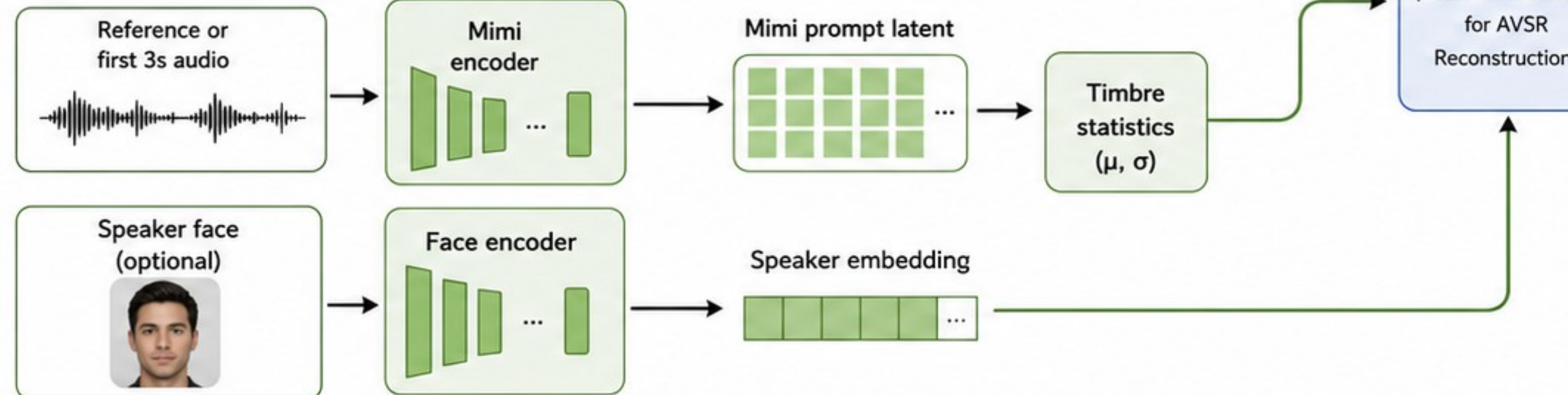
1) Visual (lip-reading) condition



2) Text (semantic) condition



3) Speaker (timbre) condition



1 CORE ARCHITECTURE

- Conditions: lip/video | Weak text hidden states | Speaker embedding | 3.04s Mimi audio prompt | Prompt timbre stats
- Target: normalized Mimi latents → waveform via Mimi decoder (offline, 24 kHz mono)
- Key: text is NOT the main generation path — lip and audio-prompt latents carry most of the signal

4 TEXT ABLATION

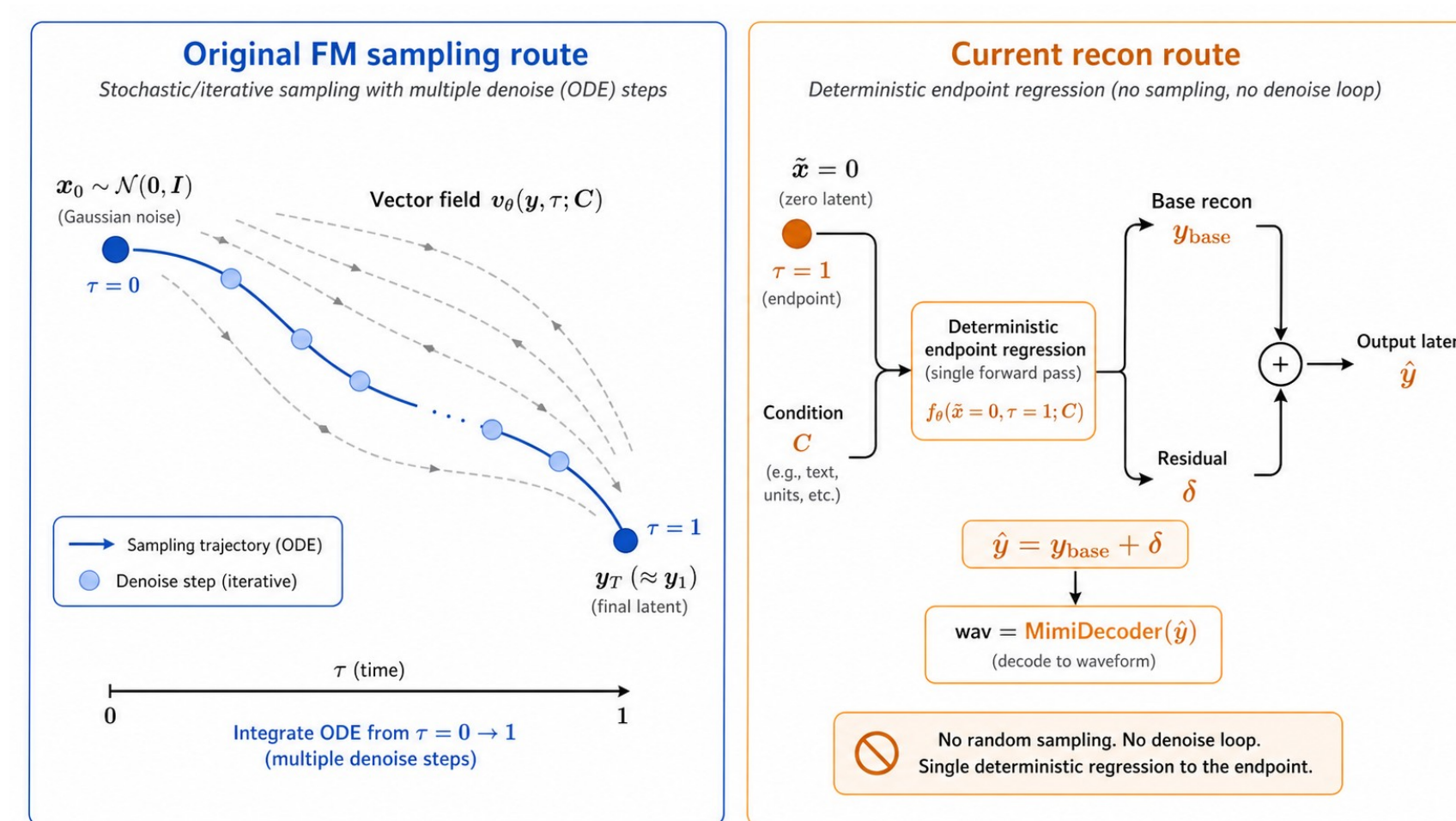
GT transcript is not the core bottleneck.

Text source	Corr	MSE	MAE
GT text + timestamps	0.5818	0.6676	0.6018
AVSR text + timestamps	0.5783	0.6716	0.6037
AVSR text + uniform	0.2868	1.684	1.024
V5 text + uniform	0.2861	1.685	1.024

- AVSR vs GT: $\Delta\text{corr} = 0.0035$ only.
- V5 (WER 29%) vs AVSR (WER 20%): $\Delta\text{corr} = 0.0007$.
- Alignment quality (timestamps vs uniform) = $\Delta 0.29$.

alignment > accuracy

5 RESIDUAL RECON



FM sampling (NFE=10) stalls at $\text{corr} \approx 0.35$.
Fix: set $\tilde{x}=0, \tau=1$ — predict endpoint directly.
• Frozen base reconstructor predicts latent.
• Trainable residual head corrects errors.
Doubles corr vs naive endpoint.
 $\hat{y} = f_{\text{base}}(C) + \Delta(C)$

2 PROJECT OVERVIEW

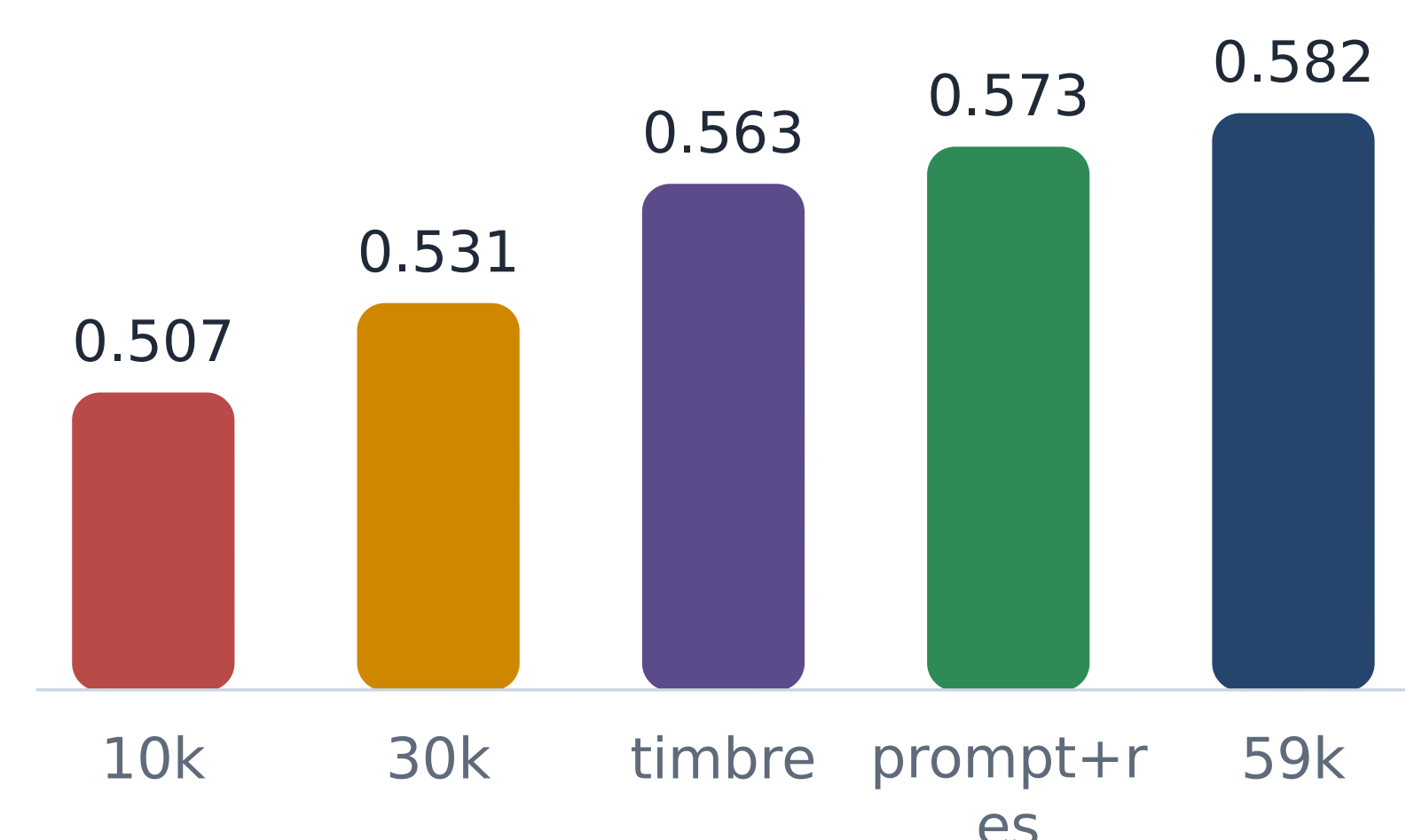
- Main Claim
Do not turn visual speech generation into a strict lip → text → TTS cascade. Text is a weak semantic/alignment cue.
- Route
Lip motion, weak LM text, speaker identity, and reference-timbre prompt are fused to reconstruct continuous Mimi audio latents.
- Evidence
Trained on 59,144 LRS3 lip-AVSR clips; final checkpoint reaches $\text{corr} 0.5818$ / $\text{MSE} 0.6676$ / $\text{MAE} 0.6018$.

3 DATA + TRAINING SIGNALS

59,144 lip-AVSR clips | fixed 1,000 val

- lip_avsr.npy — Auto-AVSR lip crop
- latentavsr_enc_lipavsr.npy — AVSR visual encoder state
- avsr_enc_lipavsr.npy — AVSR visual encoder state
- avsr_text_lipavsr.txt — noisy text, weak semantic aid
- SmolLM2 hidden — aligned text-token hidden states
- speaker/timbre — speaker emb + Mimi

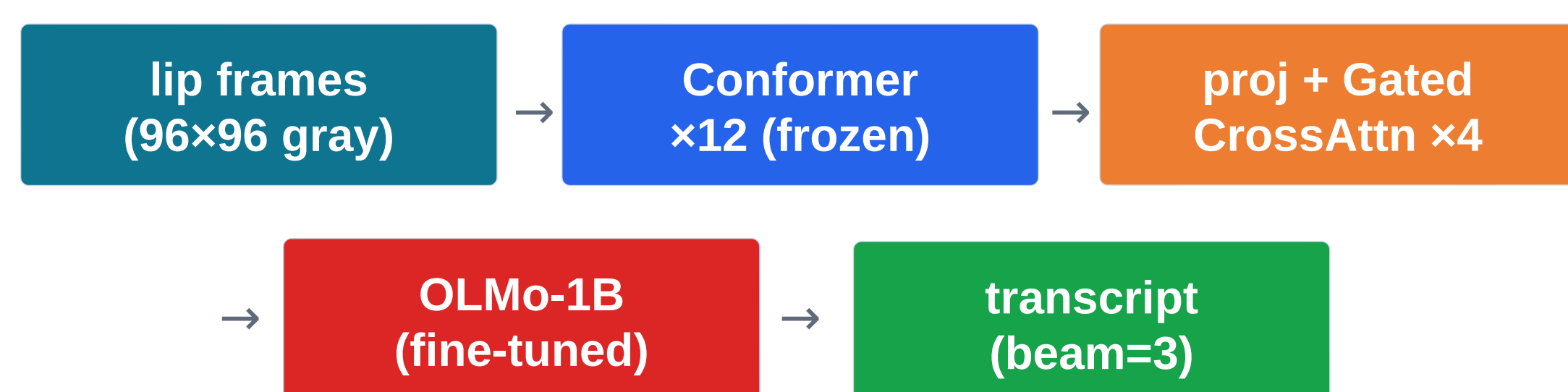
6 EXPERIMENTAL COMPARISON



Scale + timbre + residual progression (corr)
• 10k visual prior: 0.507.
• Final 59k prompt-stats residual: 0.582.
• Timbre/audio-prompt conditioning and residual recon produce the larger gains.
Final corr 0.5818 | MSE 0.6676 | MAE 0.6018

7 STREAMLIP V5: SELF-TRAINED VSR

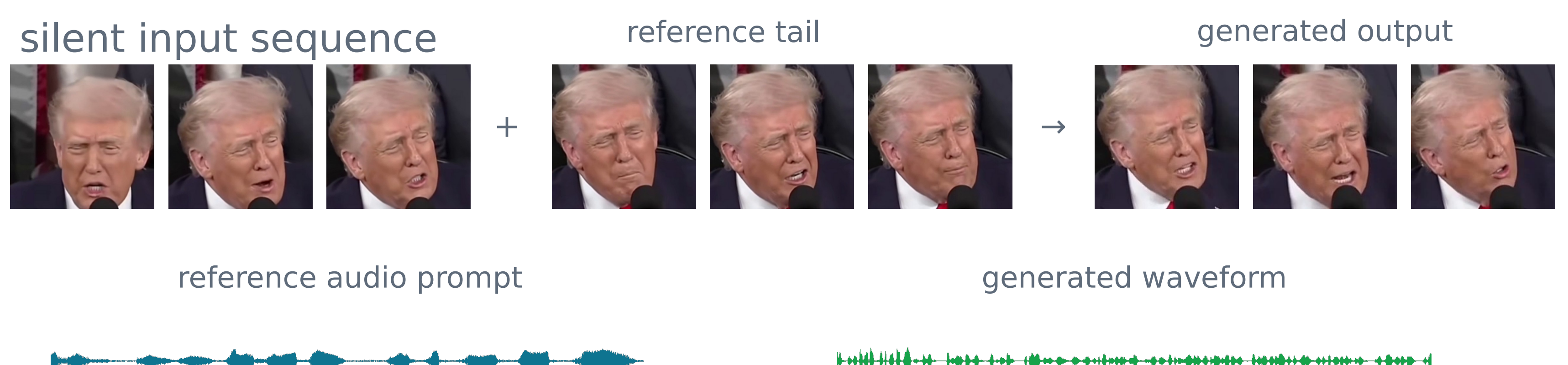
BRANCH Architecture



- Data: LRS3 pretrain split ~115k, 102k left after filtering out the blurry samples.
- Model: pretrained AV-HuBERT video conformer encoder + OLMo-1B language model. Text tokens uses Flamingo-styled Gated Cross-Attention to query the conformer KV pairs and to integrate visual information.
- Training: first fine-tune the LM on texts of samples in LRS3 pretrain split, then train the GCA weights and fine-tune the LM using standard CE loss.
- Test: test on the LRS3 test split. StreamLip V5 shows capability on long sentences, grammar correctness

System	WER	Word Acc	Notes
Auto-AVSR (beam=10)	20.2%	79.8%	CTC+Att, 5k vocab
StreamLip V5 (beam=10)	28.6%	71.4%	OLMo-1B, 50k vocab

8 SILENT → VOICED DEMO



Silent video + reference audio → AVSR text → Mimi recon → Voiced MP4

Trump corr: 0.3035 → 0.3869 with final lip-AVSR preprocessing path

9 TAKEAWAY

- Do not cascade lip → text → TTS.
- Lip motion provides phonetic and timing evidence.
- Mimi latents preserve codec-compatible acoustic structure.
- Weak text stabilizes content — does not need to be perfect.
- Timbre/reference conditions carry speaker and recording color.
- StreamLip V5: LM-based VSR can serve as self-trained text branch.

10 REPRODUCE / ARTIFACTS

- One-command raw-video demo:
python scripts/run_raw_video_avsr_recon_pipeline.py --input data/trump.mov --exp demo --force
- Gradio GUI
scripts/gradio_avsr_gui.py
- Checkpoint
Timbre-fix residual recon: step_002000.pt
Fine-tuned OLMo 1B: olmo-1b-lrs3-lr3e-5_ep2
StreamLip V5 ckpt: v5_olmo_lr1e-6_ep50_eos_frame500_warmup0/step_001500.pt